

DATA ENGINEER

AWS Data Engineer | Scalable data pipelines (3M+ records/day) | Cost optimization | Glue | S3 | Redshift | PySpark

PROFESSIONAL SUMMARY

AWS Data Engineer with 2+ years of experience building scalable ETL pipelines processing 3M+ records/day. Specialized in S3 data lakes, Glue (PySpark) transformations, and Redshift data warehousing. Improved pipeline performance by 40% and reduced AWS costs by 15% using partitioning and incremental processing.

PROFESSIONAL EXPERIENCE

Application Developer (Data Engineer – AWS Data Platform) – IBM (Oct 2023 – Present)

- Owned end-to-end data pipeline development including ingestion, transformation, storage, and orchestration on AWS.
- Designed and deployed scalable ETL pipelines processing 3M+ records/day, enabling low-latency analytics for downstream reporting systems.
- Designed optimized S3 partitioning strategy reducing query scan costs and improving Athena/Redshift performance.
- Processed 3M+ records daily using distributed Spark transformations (joins, aggregations, window functions) ensuring scalable data processing.
- Implemented incremental data loads using Glue job bookmarks, reducing redundant processing and improving pipeline efficiency.
- Optimized SQL queries using indexing, CTEs, partition pruning, and analytical functions, improving performance by 40%.
- Automated and monitored event-driven workflows using AWS Lambda, S3 triggers, and CloudWatch logs.

PROJECT – Retail Sales Analytics Data Platform on AWS

- Designed and built an end-to-end AWS data pipeline processing 3M+ records/day (CSV → S3 → Glue → Redshift), enabling scalable analytics
- Implemented partitioned S3 data lake using Parquet format, significantly improving query performance and reducing storage costs
- Designed star schema (fact + 5-dimension tables) in Redshift to support efficient analytical reporting
- Implemented incremental data processing using Glue job bookmarks, reducing redundant data scans and lowering AWS costs by 15%
- Developed event-driven workflows using AWS Lambda and S3 triggers with validation, logging, and error handling to ensure data reliability
- Orchestrated data workflows using Apache Airflow DAGs, managing task dependencies, scheduling pipelines, and improving overall workflow automation

TECHNICAL SKILLS

Data Engineering: ETL Pipelines, Data Pipelines, Data Warehousing, Data Modeling, Star Schema

AWS: S3, Glue, Redshift, Lambda, Athena, IAM, CloudWatch

Programming: Python, SQL, PySpark

Big Data: Spark, Batch Processing, Incremental Loads

Data Lake: Amazon S3 (Partitioning, Parquet Format, Data Lake Architecture)

Orchestration: Apache Airflow (DAGs, scheduling, task dependencies)

Databases: MySQL, Oracle SQL

Tools & Environment: Docker (local container setup), Git, CI/CD, Linux, Jira

Methodologies: Agile (Scrum), ETL/ELT.

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate
- AWS Certified Developer - Associate
- Microsoft Azure Fundamentals (AZ-900)

EDUCATION

Kalpataru Institute of Technology, Tiptur

Bachelor of Engineering – Electronics & Communication Engineering (2019–2023).